# Does an LLM Reranker Find New Assets Better than Popularity? A Study Plan on FAR-Trans

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**Assignment:** Session 1 — Research Question and Minimal Study Plan (LLM4Rec, Week 7) **Track:** Ranker **State:** planned; no experiment run yet

**Abstract** — FAR-Trans customers buy few assets and mostly the same Greek stocks, so recommending assets that are new to a customer is a hard exploration task in which popularity is a strong baseline. On warm customers — with history before the cut and a new purchase after it — we plan to test whether zero-shot LLM reranking of the top-30 popularity candidates, given the customer's purchase history, improves nDCG@10 over the popularity order, the classical baseline; a second iteration tests whether FLARKO's market context adds to the better of the two. In the raw data, 12 assets account for 51.7% of purchases and 60.5% of buy transactions repeat an earlier purchase, so the new-assets rule leaves only exploration. The design isolates the LLM's contribution, so even a null result tells the owner of an advisor tool whether an LLM stage is worth its latency and cost.

*Index Terms — recommender systems, large language models, reranking, financial asset recommendation, FAR-Trans*

## 1. Introduction

**Problem.** The shared project asks for a Top-10 list of financial assets for every evaluated FAR-Trans customer, scored by nDCG@10 against the assets the customer buys in a future window [1]. The data cover 29,090 customers of a European bank and 806 stocks, bonds and funds (2018–2022). The official item scope, time cut and label rule are pending; this plan assumes, as in [1], that only assets *new to the customer* count.

**Motivation.** Purchases are concentrated: stocks make 88.9% of buys, 12 assets make 51.7%, and 51.3% of customers ever bought a single asset. 60.5% of buy rows repeat an earlier purchase, so excluding repeats leaves pure exploration, the part that next-basket studies find hardest [5]. Popularity is therefore a strong baseline [1], [6]; an LLM may add knowledge about companies and sectors, but zero-shot LLM rankers show popularity and position bias [3].

**Concrete example.** The randomly drawn record of §3.1 belongs to a customer who had bought National Bank of Greece — the most-bought asset before the cut — three times. Popularity puts this stock first for most customers, but under the new-assets rule it is removed from this customer's list. *Input* of the planned prompt: the customer's earlier assets (name, category) and 30 numbered candidates (name, category, sector). *Output*: the indices of 10 candidates in ranked order, scored against the customer's new purchases in the next six months.

**Research question (iteration 1).** We test whether **zero-shot LLM reranking of the top-30 popularity candidates, given the customer's purchase history,** improves **nDCG@10** over **the same candidates in popularity order — the classical baseline of [1], which here is also the control —** for **warm FAR-Trans customers: customers with transactions before the cut who buy at least one asset new to them after it (§3.2)**. Only assets new to the customer count. No direction is assumed; a null or negative result will be reported. *Iteration 2* (planned, §3.2) adds FLARKO's market context [4] to the better of the two.

**Why this method.** (i) *Reranking, not generation.* The shared task fixes the eligible assets, and a ranker may not change them. Reordering a fixed list also makes the comparison clean: both lists hold the same assets, so a difference in nDCG@10 can only come from the order. Here we depart from [4], which generates assets freely. (ii) *Popularity as baseline and control.* It is the reference method of [1], strong on concentrated purchase data [1], [6], and it supplies the candidates, so the control is exactly the baseline's own order. (iii) *Zero-shot first.* It needs no training data or GPU and answers the basic question — does an LLM add anything over popularity — before fine-tuning as in [4] is worth its cost. (iv) *Purchase history in the prompt.* It is the only personal signal observed before the cut without leakage (profile fields are back-filled, §3.1); without it the LLM could rank only by general fame, which duplicates popularity — ablation A3 checks this. (v) *Top-30 candidates.* A reranker cannot recover a purchase that is not among its candidates [7], so the list must be clearly longer than the ten slots it fills. A longer list, however, makes the prompt slower and costlier — and runtime breaks ties in the benchmark — and strengthens position bias [3]. The LLM rankers in [3], [7] used 20 candidates; we start from 30 to raise that ceiling while keeping one prompt short, and fix the final size on validation cuts in Session 2 by checking how many future purchases fall inside the top 20, 30 and 50.

**Why this group.** We study *warm customers*: customers with transactions before the cut who buy at least one asset new to them after it. (i) The method reads the purchase history, so it is undefined for customers without one, whereas popularity works for everyone; a fair comparison is possible only where both methods are defined. (ii) nDCG@10 needs at least one relevant item, and only new assets count. (iii) The same set is used in [1], there so that different metrics are computed on the same customers; we keep it so that every team member reports on the same customers, cuts and candidates, which also keeps our results comparable with [1]. Cold customers are left out of the comparison and receive the popularity list in the team submission; because short histories dominate the data, results are also broken down by history length. Rejected alternatives: all customers (the method is undefined for cold ones), a single segment (segments come from back-filled profiles), long histories only (too few customers).

**Research importance.** The study measures whether an LLM adds ranking signal beyond popularity in a domain with very short histories, heavy concentration and exploration-only labels. Prior LLM work on FAR-Trans [4] reports top-3 hits for a fine-tuned generator, and simple tuned baselines often match neural recommenders [6]; a controlled reranking result under nDCG@10 fills that gap either way.

**Business value.** The stakeholder is the product owner of an advisor-facing recommendation tool, such as the one built on National Bank of Greece data in [2]. The decision that changes is whether to add an LLM stage — with its latency and API cost — to the advisor's shortlist or keep a popularity list. We make no claim about returns, suitability or customer acceptance.

**Contributions.** (i) A study plan with a same-information classical control and three ablations that separate the LLM's contribution. (ii) A field-level audit of FAR-Trans that fixes which inputs may be used without future leakage.

## 2. Related Work

| Study (authors, year, venue) | Method | Data and evaluation | Limitations and connection to our question |
|---|---|---|---|
| Sanz-Cruzado, Droukas, McCreadie, 2024, Fin-RecSys@IJCAI [1] | Releases FAR-Trans; benchmarks price regressors, Popularity, ARM, UB-kNN, MF, LightGCN, LightGBM hybrids | 29,090 customers, 806 assets; 61 cuts, 6-month windows, only newly acquired assets relevant; nDCG@10, ROI@10. LightGCN 0.340, Popularity 0.271 | No LLMs; tuned for ROI. Gives our data, protocol and baselines; Popularity is the bar to clear |
| Spadea, Seneviratne, 2025, ICAIF '25 [4] | Qwen3 0.6–8B with personal and market knowledge graphs generates 3 assets; LoRA + KTO fine-tuning | FAR-Trans; test 2021-12 → 2022-11; relevant = bought within 180 days (and profitable for Comb@3); Hits@3 vs baselines of [1] | Nearest prior work. Top-3 hits, generation after fine-tuning; no nDCG@10, no fixed-candidate reranking, baseline scores only in a figure |
| Hou et al., 2024, ECIR [3] | Zero-shot LLM ranks a candidate list given the user's history as text | MovieLens-1M, Amazon Games; leave-one-out; NDCG@10 of best prompt 45.8 vs trained SASRec 78.9 (ML-1M) | Position and popularity bias, weak use of history order; mitigations: recency prompt, candidate shuffling. Template for our reranker |
| Li, Jullien, Ariannezhad, de Rijke, 2023, ACM TOIS [5] | Re-evaluates next-basket methods with metrics split into repeat and explore items | TaFeng, Dunnhumby, Instacart; Recall/NDCG/PHR@K; repeat items give >97% of the best methods' performance on two datasets | Grocery, not finance. Explains why a new-assets-only label makes FAR-Trans a pure exploration task |
| Ferrari Dacrema, Cremonesi, Jannach, 2019, ACM RecSys [6] | Reproduces 18 neural top-N papers against tuned simple baselines | Original datasets; 7 of 18 reproducible, 6 of 7 often beaten by TopPop, kNN or graph random walks | Not finance or LLMs. Justifies a tuned popularity baseline and a same-information control |

**Nearest prior work and gap.** [4] is the only LLM study on FAR-Trans we found. It answered whether fine-tuned small LLMs with knowledge-graph context recommend assets that customers later buy (Pref@3) and that are also profitable (Comb@3). It left open a top-10 ranking scored by nDCG@10, an LLM that only reorders a fixed candidate list, a classical control with the same information, and an evaluation restricted to assets new to the customer. We did not find a study reporting nDCG@10 for LLM reranking of a fixed candidate set against popularity on FAR-Trans when only new assets count.

**STUDENT TO WRITE / CONFIRM IN OWN WORDS BEFORE SUBMISSION — remove this box**

Open [4] beyond the abstract and check how it treats assets the customer already held; adjust the gap sentence if needed. Open every cited paper yourself — only opened sources may be cited.

**Alternatives considered.** *Encoder track* (LLM embeddings of history or assets): deferred — asset text is thin (sector missing for all funds). *News context as RAG*: kept as hypothesis H1 for Session 3 — no dated news source for Greek companies yet. *Fine-tuning* as in [4]: optional for Session 4, not needed to answer this question.

## 3. Method

### 3.1 One actual data example

```
source: FAR-Trans public release (University of Glasgow, researchdata.gla.ac.uk/1658), transactions.csv, line
75,804;
        assetName joined from asset_information.csv by ISIN; row drawn uniformly at random (pandas, seed 2026)
record id: transactionID 10559805
fields: customerID, ISIN, transactionType, timestamp | assetName
values: 133BB7…(masked), GRS003003035, Buy, 2022-03-21 | NATIONAL BANK OF GREECE
window: history (input) side of an assumed cut T = 2022-05-30 (official split pending)
```

| Field | Status | Timestamp | Leakage check |
|---|---|---|---|
| customerID, ISIN | observed | static identifiers | identifiers only; customerID is never sent to the LLM |
| transactionType = Buy | observed | 2022-03-21 | an input only because 2022-03-21 < T; rows after T are labels, never inputs |
| timestamp | observed | 2022-03-21 | decides the side of the cut |
| assetName (+ category, sector) | observed, static metadata | asset row 2018-01-02 | no future value in the field; the LLM may know the company's later history — contamination, tested by ablation A2 |

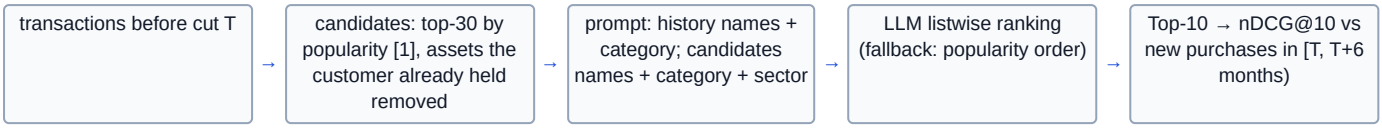
The customer had bought this asset three times before, so the row is a repeat purchase: under the new-assets rule it is history only. Not used: totalValue, units, channel (not needed); all customer-profile fields — risk level and capacity never change across snapshots and 38.5% of profile rows carry a questionnaire date later than the row date, i.e. values are back-filled; 24.5% of customers have an algorithm-predicted risk level. Our recount of Table 1 in [1] from the raw files matches every count except three that differ by one record.

### 3.2 Population, cuts and selection (common for the whole team)

**Population.** Warm customers as defined and argued in §1; results broken down by one asset, two to three, and four or more distinct assets before the cut. Cold customers receive the popularity list in the team submission.

**Cuts and selection.** Cuts follow the schedule of [1]. Earlier cuts, whose label windows end before the test period, are used for every choice; later cuts are run once for the final comparison. The LLM pilot uses one shared, seeded sample of customers. Iteration 2, which adds FLARKO's market context [4], is compared with the winner of iteration 1 in the same way.

### 3.3 Pipeline




### 3.4 Baseline, control and test design

| Element | Plan |
|---|---|
| Data and users | `transactions.csv` (customerID, ISIN, transactionType, timestamp), `asset_information.csv` (name, category, sector) and, in iteration 2, `close_prices.csv` — all observed; history = rows before T; label window [T, T+6 months). Customers, cuts and subgroups as in §3.2. |
| Baseline (= control in iteration 1) | Popularity as in [1]: assets ranked by the number of distinct customers who bought them before T, assets the customer already held removed. Further classical baselines: popularity over the last 90 days; PPR (course W6) as a stronger graph baseline. |
| Intervention | Iteration 1: zero-shot listwise LLM reranker of the baseline's top-30 given the customer's purchase history; one call per customer; output = 10 candidate indices. Iteration 2: the same prompt plus a 10-week price summary per candidate (prices before T only). |
| Fair comparison | Same customers, cuts, candidates (fixed only to isolate reranking) and metric for every method. Same-information classical controls: candidates re-sorted by overlap of category and sector with the customer's history, ties by popularity (iteration 1), and additionally by the 10-week price change (iteration 2) — the information the LLM sees, without the LLM. |
| Measure | nDCG@10 (primary); Recall@10, HR@10; candidate recall@30 (ceiling of any reranker); end-to-end runtime per customer; tokens and API cost. |
| Contribution check | A0 remove the LLM (= control); A1 shuffle candidate order before the call (position bias [3]); A2 replace asset names with anonymised "category, sector" descriptors (pre-training knowledge); A3 remove the history (does the LLM personalise or just re-rank by fame?). |
| Feasibility | S2: freeze the protocol with the team; popularity, 90-day popularity, PPR and evaluation code; sanity checks (§4); iteration 1 pilot on the shared customer sample. S3: iteration 1 on all validation cuts; iteration 2 (market context, RAG-style). S4: optional KTO fine-tuning as in [4] against the winner of iteration 2. S5: one run on the test cuts, ablations A1–A3, failure analysis. S6: report and defence. Tools: Python, pandas, Anthropic API called from the local Linux VM (host reachable; no GPU needed). Budget: pilot first, full run only if cost per cut is acceptable. |
| Variance | Temperature 0; the pilot is run three times to measure run-to-run variation; results reported as mean ± s.d. across cuts. |
| LLM and judging | Claude Haiku 5.5 (`claude-haiku-5-5`) via the Anthropic API; version and prompt frozen in Session 2. No LLM judge: ground truth is observed purchases only. |

**Key design decisions.** (1) New-assets-only labels follow [1] while the course rule is pending. (2) Reranking instead of generation, popularity as baseline and control, zero-shot first, history in the prompt and the candidate size: reasons in §1. (3) No profile fields, because they are back-filled (§3.1).

**Risks.** *Leakage*: only rows before T; profile fields excluded. *Contamination*: the LLM knows these Greek companies and possibly their 2021–2022 history, and FAR-Trans is public since 2024 — ablation A2. *Privacy*: pseudonymous IDs stay local; prompts contain only asset names and categories. *Ethics*: nDCG@10 measures purchase prediction, not investment quality or suitability; no profit claims.

## 4. Experiments (planned — nothing run yet)

| Run | Role | What it answers |
|---|---|---|
| Popularity [1] | baseline = control (iteration 1) | the bar to clear; its top-30 are the candidates |
| LLM + history, top-30 reranked | intervention (iteration 1) | the research question |
| Category/sector-overlap reranker | same-information control | is a gain due to the LLM or to the fields it sees? |
| 90-day popularity; PPR | further baselines | does the LLM beat stronger classical lists, not only its own candidates? |
| A1 shuffled order; A2 anonymised names; A3 no history | ablations | position bias; pre-training knowledge; personalisation |
| LLM + history + 10-week price summary [4]; popularity re-sorted by 10-week price change | iteration 2: intervention; same-information control | does FLARKO's market context add to the winner of iteration 1? |

**How the hypothesis is tested.** For every cut, nDCG@10 is computed per customer for the LLM list and for the control on the same customers. We report the mean over customers and cuts and the paired difference (LLM − control) with a 95% bootstrap interval over customers; an interval that contains zero is reported as a null result. The same paired comparison is made against the same-information control and against 90-day popularity and PPR, and repeated within the history-length subgroups. Ablations are read as follows: a gain that shrinks under A1 points to position bias, one that disappears under A2 came from knowledge about the companies, and no drop under A3 means the LLM is not personalising. Every choice — prompt, K, the winner of each iteration — is made on validation cuts; the test cuts are run once, and every iteration is reported, including failed ones.

**Planned verification.** (a) Re-implemented popularity must reproduce the nDCG@10 reported in [1] (0.271) up to differences in cut dates. (b) nDCG@10 is computed by hand for one customer and compared with the code output. (c) Every LLM output is parsed and checked to contain 10 distinct candidate indices; failures fall back to the popularity order and are counted.

## 5. Discussion

### Team discussion

**Team proposal (Zlata Kranina, 12 November 2026).** Zlata proposed one common theme — LLMs for asset recommendation on FAR-Trans — with a separate angle, literature, question and experiment for each of the three members: (1) short histories, LLM as encoder: CF plus LLM-based similarity of assets vs popularity and CF, for customers with at most three distinct assets; (2) fit with the risk profile, LLM as ranker: LLM reranking of the popularity top-20 with the customer's risk profile vs the same top-20 without the LLM, scored by nDCG@10 and the share of risk-suitable recommendations; (3) asset descriptions: features extracted by an LLM from asset names (type, sector, risk level) vs the original category fields in a content-based recommender, for customers who bought assets with an empty sector. The common part: team name and leader, a repository with the instructor as collaborator, CONTRIBUTIONS.md and team_notes.md, and one evaluation scheme — what counts as a correct answer, the time split, nDCG@10, the candidate assets, a rule for transaction IDs shared by several customers, and popularity and simple CF as non-LLM baselines. A second meeting records agreements and open questions. Two members may take the same slot.

**Options I brought.** I prepared the options below with the AI assistant (session log) and brought them to the team discussion; the second column is my proposal, not a team decision.

| Option | Proposal | Reason |
|---|---|---|
| **Methods and questions** | | |
| LLM + purchase history reranks the popularity top-30, vs popularity order | chosen: iteration 1 | one change; only data observed before the cut; feasible via API |
| FLARKO-style generation of assets [4] | rejected | a ranker may not change the eligible assets; generation breaks the fixed-candidate comparison |
| FLARKO market context: 10-week price summary per candidate [4] | iteration 2 | uses only FAR-Trans prices; tests one component of [4] |
| FLARKO KTO fine-tuning [4] | optional, Session 4 | needs a GPU and an open model |
| LLM vs a LightGBM learned ranker with the same features | possible later control | needs feature engineering; the LambdaMART hybrid of [1] did not beat popularity |
| Real asset names vs anonymised descriptors | ablation A2 | measures pre-training knowledge; does not raise the benchmark by itself |
| History vs no history in the prompt | ablation A3 | shows whether the LLM personalises |
| Shuffled candidate order, several passes | ablation A1, one shuffle | several passes multiply calls and runtime; already studied in [3] |
| Last five purchases vs full history | dropped | few long histories; repeats a finding of [3] |
| Encoder track: LLM embeddings of histories or asset descriptions | dropped for now | open embedding models are not reachable from our environment; asset text is thin |
| News headlines as RAG context (hypothesis H1) | deferred to Session 3 | no dated news source for Greek companies yet; leakage and contamination risks |
| Encoder + reranker hybrid | dropped | highest effort: two LLM components and four runs |
| Baselines: 90-day popularity, PPR / MF, user-kNN | kept / dropped | stronger classical references / weaker than popularity in [1] |
| Candidate list of 20, 30 or 50 | 30, provisional | fixed on validation cuts in Session 2 (§1) |
| **Customer groups** | | |
| Warm customers | chosen | reasons in §1 |
| Cold customers | excluded | the method is undefined; popularity fallback in the team submission |
| Short histories only | breakdown, not the population | reported for every method |
| Long histories only | rejected | too few customers |
| A single customer segment | rejected | segments come from back-filled profiles |
| Time periods | used as the split | validation vs test cuts (§3.2), not a customer group |

**My position and reasons.** I take the ranker slot with my own intervention: the LLM reranks the popularity candidates using the customer's purchase history, and iteration 2 adds FLARKO's market context [4]. The purchase history is observed before the cut without leakage, whereas the risk profile in FAR-Trans is back-filled and partly predicted by an algorithm (§3.1); FLARKO is the nearest prior work on the same data and gives the second iteration a clear basis. If another member also takes the ranker slot with the risk profile, we share candidates, prompt skeleton, answer parser and evaluation code and differ in what the prompt adds; shared code is marked as joint work.

**My contribution and the points I raise for the team.** (i) One population for all members — warm customers — so that our nDCG@10 values are comparable (§1, §3.2). (ii) Candidates defined only from data before the cut: "assets traded in the test period" would reveal future purchases. (iii) A decision on repeat purchases: under the new-assets rule a "latest buy" baseline recommends nothing new; the instructor's answer is pending. (iv) A shared transaction ID counted once for popularity and kept in each customer's history. (v) Data facts for the other angles: the risk profile never changes across snapshots and is partly predicted;

the sector is missing for all funds. (vi) One evaluation script, one pilot sample and one candidate-list size for the team. (vii) The news-attention hypothesis H1 for RAG and the scheme in which the winner of one iteration becomes the control of the next. *Open:* the candidate-list size (20 in Zlata's plan, 30 here), the repeat rule and the evaluated users — to be fixed at the second meeting or by the course.

### Failure case, root cause, fix

No model was run in Session 1, so the failure case comes from the data analysis. **Failure:** the AI assistant first stated that customers mostly buy the same mutual funds. **Root cause:** it generalised from the catalogue, where funds are the largest asset group, to purchases. **Fix and verification:** buy shares by asset category were computed from `transactions.csv` joined with `asset_information.csv`; stocks dominate purchases and funds are a small share, so the claim was corrected before it entered the design. A second check of the same kind: the customer profile looked usable until questionnaire dates were compared with snapshot dates, which showed back-filled values; the profile was excluded from the inputs (§3.1).

**Threats to validity.** The official protocol is pending (cut, item scope, repeat rule); 35,686 buys are dated 2018-01-02, likely an opening-portfolio snapshot; 23,388 transaction IDs are shared by 2–5 customers (likely joint accounts) and inflate popularity counts; one bank, mostly Greek stocks, and a COVID-era regime limit generality; pre-training contamination may favour the LLM.

**Meaning of results.** A gain over the control and over the same-information control would mean the LLM adds signal beyond popularity and simple content overlap; if the gain vanishes under A2, it came from knowledge about the companies rather than the customer's history. A null or negative result would be consistent with [5], [6] and would support keeping the cheaper popularity or graph list.

**What worked.** Auditing the raw files before choosing the question, which exposed the back-filled profile and the shared transaction IDs; tying every method to a source; separating baseline, control and same-information control. **What surprised me.** Purchases go mostly to Greek stocks, not to the funds that dominate the catalogue; profile fields are filled in after the fact; FLARKO is the only LLM study on FAR-Trans we found. **Next improvement.** Fix the label rule and the candidate-list size with the team and the instructor, then run the popularity sanity check and the iteration-1 pilot in Session 2.

## 6. AI Usage Disclosure

**Tools.** Claude (Anthropic), agent session in the Claude app with access to the local course folder; configured model `claude-opus-5-5`. Python 3.10.12, pandas 2.3.3, numpy 2.2.6 for the scripts it wrote and ran.

**How AI was used.** *Search*: literature candidates and paper details. *Code*: data analysis and leakage audit, record selection and the data dashboard (`session1_work/*.py`, `fartrans_overview.html`); a popularity-baseline script was also written and run as a check, and its results are not used in this report. *Docs*: summaries of the assignment, the related-work table and this draft. *Viz*: the data dashboard.

**What I personally verified.** I opened the cited papers myself and read them beyond the abstract. I inspected the raw FAR-Trans files myself. Together with the AI assistant, as the session log shows, I re-checked the methods and the customer groups: I asked for the source behind each method, questioned the roles of baseline and control, removed experiment numbers that were not part of this assignment, and checked the dataset paper's own reason for keeping warm customers. **What I trusted without verification.** The descriptive statistics produced by the AI-written scripts (data audit and dashboard) were not recomputed by hand. **Session log.** `vladislav_kutranov_session1_session.json`.

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